

URBAN IBILE

Updated Project Requirements & Current Source of Truth

Backend, Database, Admin, Website & Mobile App

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Prepared for the Urban Ibile Development Team

Source-of-truth rule: Where an older feature brief, repository note or previous requirement conflicts with this document, this document should be followed. The project is not being restarted; existing work should be retained unless it conflicts with the updated requirements.

Important: This is the current structural update, not a promise that the database will never need another migration. Tailor fitting information and the Jumia Logistics arrangement are still pending and may cause smaller updates later.

1. Document Purpose and Status

This document brings the latest client clarifications, approved features, development decisions and pending items into one place so the whole team works from the same information.

The goal is to update the existing schema and backend without throwing away completed work. Most existing authentication, payment, stock, order and infrastructure foundations can remain.

Requirement status labels

CONFIRMED	Client requirement or agreed technical direction that should be implemented now.
PENDING	Information expected later. Structure should remain flexible until it arrives.
TEAM DESIGN	A product/development decision we are responsible for designing, then confirming where required.
REPLACES OLD REQUIREMENT	This item overrides a conflicting requirement in the older brief.

What this update changes most

- Blueprint / Find My Size is now based on Height + Weight + Fit Preference rather than the older chest/waist/hip structure.
- A proper Return / Exchange backend structure is required and must retain Blueprint history for Fault-Line decisions.
- The secret-page requirement changed from 14 QR codes / 14 pages to 8 permanent QR codes / 8 secret pages.
- The client selected the more protected permanent-QR option (Option 2).
- Secret QR-page content is not editable from the admin panel; placeholders are used now and developers will replace them when final content arrives.
- The tailor will provide the actual Blueprint fitting information next week.
- The client will confirm the Jumia Logistics arrangement next week.
- The customer website framework is still undecided; do not treat Next.js as final.

2. Overall Platform Architecture

Status: CONFIRMED

Urban Ibile will have a customer website, customer mobile application, one shared admin panel, one shared backend and one shared PostgreSQL database.

The website and mobile app must use the same products, prices, sizes, stock, customer records, orders and payment information. The admin panel also works with the same backend and database.

Example:

Product A / Medium stock = 5
 A mobile-app customer buys 1
 Website + admin must now see Medium stock = 4

Do not create separate inventory or order systems for web and mobile.

Current backend technology

- Node.js
- Express.js
- TypeScript
- PostgreSQL
- Kysely
- pg

The existing supporting tools can remain: Zod, Pino, Helmet, JWT/Jose, Argon2, CORS and rate limiting. There is no reason to restart the backend or switch the current database layer to Prisma.

Redis/BullMQ are not required immediately. They can be added later if background queues, retries, scheduled work, email processing, video processing or similar jobs genuinely require them.

Customer website frontend

Status: PENDING - React and Next.js have been discussed, but neither is the final approved customer website framework yet. Backend/database work should not depend on this decision.

3. Customer Accounts and Admin Access

Customer accounts

Status: CONFIRMED

Customers must create an account and sign in before completing a purchase. The customer account should support the following:

- Register, login and logout
- Email verification
- Forgot/reset password
- Basic profile management
- Delivery addresses/details
- Order history
- Viewing an individual previous order
- Starting a Return / Exchange request from a purchased item

The existing customer authentication foundation can remain.

Admin users

There will be one shared admin system for both the website and mobile app. The company may have one or two authorised administrators, and each admin should have their own login. Existing admin authentication can remain.

Important operational/admin actions should continue to be auditable where appropriate.

4. Product Catalogue, Search, Video and Language

Products and inventory

Products should support the information required for the store, including:

- Product name
- Clothing type/category
- Description
- Price
- Product video
- Available sizes / variants
- Stock per size
- Product status
- Any garment/cut information later needed by Blueprint

Stock must be tracked per product-size variant, not only as one total number for the whole product.

```
Urban Shirt
Small = 5
Medium = 8
Large = 3
XL = 0
```

Shop pagination

The store should support up to approximately 100 product video listings initially. The approved pagination is 20 outfits per page, giving up to 5 populated pages when all 100 products exist. Empty pages should not be shown.

Because the shop is video-heavy, the frontend should avoid loading all 100 product videos at once.

Product search

Customers must be able to search the full catalogue by outfit/product name and clothing type (for example shirt, trouser or jacket). Search must not be limited to the current pagination page.

Search results should show the product video, product name and price. A no-result state should use the appropriate English/Pidgin copy according to the selected language.

Video behaviour

- Main homepage/background fashion video automatically plays/repeats.
- Shop/product listings use moving model/product videos.

- Product-detail video repeats automatically and should not show normal playback controls in the final experience.
- Videos visible on screen may play together; off-screen videos should pause where appropriate.
- Temporary videos are used during development because final client videos are pending.
- Final homepage and product videos must be replaceable later.
- Large video files are not stored directly in PostgreSQL. PostgreSQL stores media records/locations; the actual files live in media/object storage.

Final video specification

Status: TEAM DESIGN - the development team must decide suitable resolution, format, encoding, compression, file size, length and aspect ratio, balancing visual quality with performance. These specifications will then be given to the client.

English / Pidgin toggle

English is the default. The client has a professional writer who will provide approved Pidgin copy. Translation is not automatic.

```

English copy prepared
↓
Client/writer translates
↓
Approved Pidgin returned
↓
Both approved versions are added

```

The language switch should preserve the customer's current page/cart/context where practical.

Product names, prices/currency, search behaviour and Blueprint sizing logic remain unchanged by the language toggle. The admin panel does not require a full translation-management section. Final Pidgin copy is still pending.

5. Shared Admin Panel

The shared admin panel should eventually support normal e-commerce operations, including:

- Products and descriptions
- Prices
- Sizes/variants
- Stock
- Product videos
- Homepage/background video
- Orders
- Order processing status
- Delivery information where applicable

- Other operational settings

Any current mock dashboard figures should eventually be replaced with real database data.

Order processing status

Payment status and fulfilment status must remain separate. Current approved order-processing statuses are:

- BEING_PREPARED
- SENT_OUT
- DELIVERED

Payment Status: PAID

Order Status: BEING_PREPARED

6. The Blueprint / Find My Size - Latest Requirement

Replaces old requirement: The older chest/waist/hip structure should not be treated as the current main Blueprint flow.

The latest client requirement uses three customer inputs:

- Height
- Weight
- Fit Preference

Fit Preference options are exactly: Tailored, Standard and Oversized.

Phase 1 - Preventative Lock

When the customer clicks Add to Cart, Blueprint should intercept before the item is finally added:

```

Choose outfit
↓
Add to Cart
↓
Blueprint opens
↓
Height + Weight + Fit Preference
↓
Size recommendation
↓
LOCK IN SIZE / I KNOW MY SIZE, LET ME CHOOSE
  
```

Example result: “The Blueprint recommends: Size Large (Oversized Fit).”

Information still coming from the tailor

Status: PENDING - client will confirm the actual fitting information with the tailor next week.

We already know the inputs and output shape. What we do not yet have is the real Urban Ibile mapping that tells the system which size to recommend for each combination.

Example only - NOT real brand data:

Height: 170 cm

Weight: 65 kg

Fit: Standard

→ Medium

We must not invent these mappings. The tailor information must also clarify whether different garment cuts/products can produce different recommendations for the same person.

The schema should support the final fitting information now, so receiving the tailor data becomes mainly a data/configuration update rather than a database redesign.

Blueprint decision history

Each relevant purchase should preserve enough information to know:

- Customer
- Product / garment cut where applicable
- Height entered
- Weight entered
- Fit Preference
- Recommended size
- Final selected size
- How the final size was selected
- Blueprint rule/version used
- Date/time

Selection modes

The system needs to distinguish these situations:

RECOMMENDED / LOCKED: Blueprint recommended a size and the customer accepted that same size.

OVERRIDE: Blueprint recommended one size but the customer deliberately selected another.

MANUAL: The customer used “I know my size, let me choose” and selected their own size.

The fitting decision must stay connected to the cart/order process so it is available later during Fault-Line return/exchange review.

Rule versioning / historical accuracy

If the brand changes its sizing information later, old orders must still show what Blueprint recommended at the time of purchase. Historical orders must not be silently recalculated with a newer rule set.

January: Rule Version 1 → recommends Large

March: brand updates sizing logic
January order must still show Version 1 → Large

7. Blueprint Phase 2 - Fault-Line and Return Responsibility

The client has already defined the main wrong-fit responsibility logic. The system compares the Blueprint recommendation with the size actually purchased.

Scenario A - Customer overrides Blueprint

Blueprint recommended: Large
Customer selected: Medium
Customer later says: "It does not fit."
Classification: CUSTOMER_FAULT

The customer ignored the recommendation. Under the client's rule, the customer is responsible for the return/reverse-dispatch cost. The system must be able to identify that an override occurred.

Scenario B - Blueprint fails

Blueprint recommended: Large
Customer purchased: Large
The garment still genuinely does not fit
Classification: BRAND_FAULT / BLUEPRINT_FAILURE

The brand handles the appropriate return/replacement cost. These cases should also be useful for improving the Blueprint sizing information later.

Scenario C - Change of mind

Correct size was purchased
No product defect
Customer simply does not like the appearance
Classification: CUSTOMER_FAULT

The customer is responsible for reverse-logistics cost.

8. Return / Exchange Feature

Status: Required backend structure. The client has confirmed the core Fault-Line responsibility logic; the development team still needs to finalise some operational policy details.

The customer flow should be simple:

```

My Orders
↓
Open Order
↓
Select the exact purchased item
↓

```

Return / Exchange

A request must be linked to the specific order item, because one order may contain several products and only one item may be returned or exchanged.

Return / Exchange reasons

- Does not fit
- Wrong item delivered
- Item arrived damaged/faulty
- Changed my mind
- Other

The customer should also be able to provide a short explanation.

Photo evidence

Current development design:

- Damaged/faulty item - request photo evidence.
- Wrong item delivered - request photo evidence.
- Fit complaint - photo can be optional supporting evidence.
- Change of mind - photo does not need to be required.

Uploaded images should use the existing media/object-storage system. The return record links to media records rather than storing raw image binaries directly in PostgreSQL.

Admin review information

During review, the admin should be able to see:

- Customer
- Order
- Exact order item
- Product
- Purchased size
- Return or Exchange request type
- Reason
- Customer explanation
- Uploaded evidence where applicable
- Blueprint recommended size
- Actual selected size
- Whether Blueprint was accepted or overridden
- Fault classification
- Request status
- Admin decision
- Admin notes
- Review timestamps

Responsibility states

- CUSTOMER_FAULT
- BRAND_FAULT
- REVIEW_REQUIRED

Not every case should be auto-decided. For example, a customer who followed Blueprint but reports a poor fit may require review before the final classification.

Operational return policy still to be finalised by the team

Status: TEAM DESIGN - do not hard-code arbitrary values yet.

The client gave the Fault-Line logic, but we still need to design and present the remaining operational policy, including:

- Return/exchange request window after delivery
- Required item condition
- Original tags requirement
- Unworn/unwashed requirement
- Exact exchange process
- Exact refund process where applicable
- Any special product restrictions
- How reverse-logistics payment is handled operationally

The database/backend should be flexible enough to apply the final policy once it is agreed.

9. Cart, Inventory, Checkout, Payments and Orders

Cart

A cart item should reference the correct product/variant and connect to the Blueprint fitting decision where applicable. Customers should be able to view the cart, change quantity, remove an item and review the selected size.

Pricing must be verified by the backend. The backend must not trust a price sent by the website/mobile app as authoritative.

Stock reservation / overselling protection

The existing stock-reservation concept should remain so two customers cannot both successfully buy the same final unit.

Only 1 Medium shirt remains
 Customer A begins checkout
 Customer B begins checkout
 → Both must not successfully purchase the same final unit

If checkout/payment is abandoned or expires, reserved stock should eventually be released.

Checkout

Checkout collects the required customer contact/delivery information and shows Product Total + Delivery Cost = Final Total before payment. The final delivery implementation remains pending Jumia confirmation.

Paystack

Status: CONFIRMED payment provider

Starting a Paystack transaction does not mean the order is paid. The backend should mark payment successful only after proper verification/confirmation.

Duplicate Paystack/network events must not create duplicate orders, deduct stock twice, confirm payment twice or repeat the same fulfilment action. Existing webhook/idempotency planning should remain.

A failed payment should not produce a successful-order state. The customer should be able to retry where appropriate without unnecessarily rebuilding the cart.

Orders and historical snapshots

After confirmed payment, the order confirmation should include:

- Order number
- Purchased products
- Selected sizes
- Quantities
- Product total
- Delivery cost
- Final amount paid
- Delivery address

The same order must appear consistently in customer order history, the admin panel and the confirmation email.

Historical orders must preserve what was purchased at that time. If a product price or product information changes later, the old order must still show the original purchase values. The same principle applies to historical Blueprint data.

Order email

A successful order should trigger an email confirmation tied to the same order record/order number. Email processing can later move to a background queue if needed, but this should not delay current schema work.

10. Delivery / Jumia Logistics

Status: PENDING - client expects to confirm the Jumia Logistics arrangement next week.

Do not build a final Jumia integration from assumptions. Keep the existing delivery structure flexible enough to store normal delivery information while we wait.

When the client confirms the arrangement, we need clarity on:

- Whether Jumia Logistics is definitely the provider
- Whether fulfilment is manual or API-based
- How orders are passed to Jumia
- Coverage areas
- Delivery fees / fee calculation
- Expected delivery times
- Tracking information
- Delivery status information
- Any API/integration requirements if applicable

Depending on the answer, there may be a smaller delivery-related backend or schema update. Do not consider the delivery part permanently finished yet.

11. Secret QR Pages - Latest Client Decision

Replaces old requirement: The previous 14 QR codes / 14 secret pages requirement is obsolete. The current requirement is 8 permanent QR codes / 8 secret website pages.

Eight permanent QR codes and eight secret pages

QR 1 → Secret Page 1
 QR 2 → Secret Page 2
 QR 3 → Secret Page 3
 QR 4 → Secret Page 4
 QR 5 → Secret Page 5
 QR 6 → Secret Page 6
 QR 7 → Secret Page 7
 QR 8 → Secret Page 8

Each QR opens only its own corresponding page. The current project needs eight records/pages, but the database schema should not be unnecessarily hard-coded so rigidly that a future business change requires rebuilding the whole feature.

Secret-page discovery rules

These pages must not be visible through normal website/app browsing. They should not appear in:

- Main navigation
- Homepage
- Shop
- Product search
- Customer account navigation
- Mobile-app navigation
- A public list of secret pages
- Normal internal links that expose all secret pages
- Ordinary search-engine discovery where avoidable

Search-engine indexing controls are only an additional precaution; they are not the security mechanism.

Secret-page content

Status: PENDING CONTENT - use placeholders now.

The client has not yet provided the final content for the eight pages. During development, use placeholders. Later the company will provide the eight final write-ups and the developers will replace the placeholders.

The secret-page content is NOT editable from the admin panel. Any older requirement saying admins manage these pages should be removed.

Permanent links and QR handover

Each secret page needs a stable internal location so its QR remains valid. Updating placeholder content later must not require reprinting or changing the QR code.

At final handover, the company should receive eight clearly labelled QR-code files, one for each secret page. The company intends to print the QR codes on packaging/boxes.

Selected security option - Option 2

Status: CONFIRMED by client

The client chose the more protected permanent-QR architecture. Functional requirement:

- Exactly 8 permanent QR codes remain printable and reusable on packaging.
- The eight secret pages remain hidden from ordinary browsing.
- A valid QR access flow grants access to its matching secret page.
- Copying the normal page URL after opening it should not automatically grant another browser/person access.
- This requires additional backend access-control work beyond simply creating hidden public URLs.

- The permanent QR itself is still shareable because anyone who possesses a copy of the valid QR can scan it.

Accepted QR limitation

The client understands that with only eight permanent QR codes printed on packaging, the system cannot reliably distinguish a person holding the original physical box from a person who received a photograph/screenshot of that same valid QR code.

The client does not want the stricter unique/expiring architecture because it does not suit packaging/printing. Therefore we are NOT using:

- One-time-use QR codes
- Expiring QR codes
- Different QR for every order
- Different QR for every customer
- Account-specific QR codes

QR backend/database implications

Update current QR-related work to reflect:

- Current requirement = 8 secret pages
- Current requirement = 8 permanent QR codes
- Stable QR-to-page association
- Protected permanent access (Option 2)
- Secret-page content not managed through admin
- Placeholder content now; final content inserted by developers later

12. Performance and Media Infrastructure

Status: CONFIRMED target: 5,000 active users at the same time across website + mobile app combined.

Example:

3,000 website users
+ 2,000 mobile users
= 5,000 active users

This does not mean PostgreSQL should receive 5,000 direct database connections. Use appropriate connection pooling and efficient backend/database behaviour.

Important areas to load/performance test before final production delivery include:

- Product browsing
- Video-heavy shop usage
- Search
- Authentication

- Cart
- Stock reservation
- Checkout
- Payment confirmation
- Order creation
- Concurrent purchasing

Large media should use the selected object/media storage and delivery infrastructure rather than forcing the Express API to stream every video directly.

Akshay is currently researching the Cloudflare/S3-compatible storage side. That can continue separately and should not block the immediate schema updates in this document.

13. Pending Information and What Happens Later

Expected next week

Blueprint fitting information from the tailor:

- Actual Height + Weight + Fit Preference → Size mappings.
- Any garment/cut-specific differences.
- This should mostly become a data/configuration update if the schema is designed flexibly now.

Jumia Logistics arrangement:

- Provider confirmation, workflow, fees, coverage, timing, tracking and integration method.
- This may require a smaller delivery backend/schema update depending on what is confirmed.

Still pending from client/company

- Final homepage/background video
- Final product/outfit videos
- Final content/write-ups for the 8 secret pages
- Final approved Pidgin copy
- Brand support email
- Phone/WhatsApp details
- Social-media links

Most of these should be content/configuration updates rather than major database redesigns.

Still to be designed/finalised by the development team

- Full operational Return / Exchange policy (request window, item condition, tags, exact exchange/refund flow, etc.)
- Final technical video specification
- Any final delivery implementation after Jumia confirmation

14. Database / Backend Work to Do Now

Keep the existing database foundation. The important current structural updates are:

A. Blueprint

- Height
- Weight
- Fit Preference
- Recommended Size
- Final Selected Size
- RECOMMENDED/LOCKED vs OVERRIDE vs MANUAL
- Rule/version history
- Product/garment-cut relationship where required
- Connection from fitting decision to cart/order history

B. Return / Exchange

- Return/Exchange request
- Customer
- Order
- Exact order item
- Reason
- Explanation
- Evidence/media links
- Blueprint decision connection
- Fault classification
- Request status
- Admin review/decision
- Admin notes
- Timestamps

C. QR

- Replace any 14-page assumption with current 8-page requirement
- 8 permanent QR codes / 8 secret pages
- Stable QR-to-page mapping
- Option 2 protected permanent access
- No admin editing for secret-page content
- Placeholder content now; developer-updated final content later

D. Delivery

Keep delivery structure flexible while waiting for the Jumia Logistics confirmation. Do not invent a final provider workflow yet.

15. Important: Do Not Treat Today's Schema Update as Permanently Final

Important: This update gives the correct current structure. It does not mean no future migration will ever be required.

When the tailor fitting information and Jumia Logistics details arrive, there may be smaller database/backend updates. The same applies if pending business information reveals a field or relationship that could not reasonably be known today.

```

Build the correct flexible structure now
    ↓
Receive pending information
    ↓
Insert/configure final values
    ↓
Only make small structural changes if genuinely necessary
  
```

Items that should mostly be data/content updates later

- Tailor's actual fitting combinations
- Final QR-page write-ups
- Final Pidgin content
- Final videos
- Support email
- Phone/WhatsApp
- Social links

Items that may still require smaller backend/schema adjustments

- Jumia Logistics final arrangement
- Tailor fitting information if garment-cut behaviour is more complex than currently expected

16. Old vs Current Requirement - Quick Reference

Area	Older / outdated assumption	Current requirement
Secret pages	14 QR codes / 14 pages	8 permanent QR codes / 8 secret pages
QR access	Hidden pages only	Option 2: permanent QR + additional backend access control; copied normal page URL should not grant access
QR content	Admin-managed	Not admin-managed; placeholders now, developers update when final content arrives
Blueprint inputs	Chest / waist / hip style structure	Height + Weight + Fit Preference

Area	Older / outdated assumption	Current requirement
Blueprint history	Not fully defined	Persist recommendation, final size, selection mode and rule/version history
Returns	Basic / incomplete	Proper Return / Exchange structure linked to order item + Blueprint + evidence + fault classification
Customer website framework	Next.js treated as decided	Still pending; do not lock backend to Next.js
Delivery	Generic delivery structure	Keep flexible; Jumia Logistics arrangement pending next week

17. Development Approach Going Forward

Do not restart working code. The immediate sequence should be:

1. Update schema for the confirmed structural changes in this document.
2. Review the updated tables together.
3. After the updated schema is ready, restore/run the current project/database locally.
4. Continue backend modules using the updated structure.
5. Add the pending client/tailor/Jumia information when received.
6. Make only the smaller follow-up updates genuinely required by that new information.

18. Current Status Summary

Category	Status	Items
Confirmed now	CONFIRMED	Core stack and shared backend/database; customer accounts; Paystack; product/video catalogue and search; English/Pidgin process; latest Blueprint and Fault-Line logic; Return/Exchange structure; 8 permanent protected QR pages; 5,000-active-user target.
Expected next week	PENDING	Tailor's actual Blueprint fitting information; Jumia Logistics arrangement.
Pending later	PENDING	Final QR-page content, videos, Pidgin copy and brand contact/social details.
Team to finalise	TEAM DESIGN	Operational Return/Exchange policy; technical video specification; final delivery implementation after Jumia confirmation.